

MASSACHUSETTS GENERAL HOSPITAL

Analytics Report

Patient Encounters, Treatment Costs & Behavioral Patterns | 2011 – 2022

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Tools: Microsoft Excel | Python | MySQL | Power BI

Dataset: Synthea Synthetic EHR — Massachusetts General Hospital

1. Project Background

This report presents the findings from an end-to-end analytics project conducted on a synthetic Electronic Health Record (EHR) dataset for Massachusetts General Hospital (MGH), Boston. The dataset was generated using Synthea, an open-source synthetic patient generator, and covers 11 years of hospital activity from January 2011 to February 2022.

The analysis was structured around three objectives — understanding encounter patterns, examining cost and payer coverage, and profiling patient demographics and behavior. The full workflow spans data profiling in Excel, exploratory analysis in Python, structured querying in MySQL, and an interactive dashboard in Power BI.

Note: All cost figures in this dataset are synthetically generated and do not represent real-world hospital billing rates. The analytical relationships and patterns observed are valid for study purposes.

Dataset Snapshot

Table	Records	Description
encounters	27,891	Patient visits with cost, payer, and duration data
patients	974	Demographics — age, gender, marital status
payers	10	Insurance payers including NO_INSURANCE category
procedures	47,701	Procedure records linked to each encounter
organizations	1	Massachusetts General Hospital, Boston MA

2. Encounters Overview

The dataset spans 27,891 patient encounters across 11 years. Encounter volume showed a significant peak in 2014 at 3,885 visits, followed by a gradual decline through 2019, and a secondary peak in 2021 at 3,500 visits. The sharp drop in 2022 reflects partial-year data, as the dataset only extends to February of that year.

Encounter Class Distribution

Ambulatory encounters dominate the dataset, accounting for nearly half of all visits. This is consistent with an outpatient-heavy facility profile. Emergency and inpatient visits represent a smaller portion but carry significantly higher claim costs.

Encounter Class	Count	% of Total
Ambulatory	12,537	44.9%
Outpatient	6,300	22.6%
Urgent Care	3,666	13.1%
Emergency	2,322	8.3%
Wellness	1,931	6.9%
Inpatient	1,135	4.1%

Encounter Duration

95.9% of all encounters lasted under 24 hours — 26,739 out of 27,891. The 1,152 encounters exceeding 24 hours are primarily inpatient and ICU admissions. These longer encounters carry a disproportionately higher average claim cost compared to same-day visits, making duration a useful predictor of cost intensity.

3. Cost & Coverage Insights

Payer Coverage Gap

One of the most significant findings in this dataset is the extent of zero payer coverage. 13,586 encounters — representing 48.7% of the total — had no insurance coverage at all. This means nearly half of all hospital visits resulted in full out-of-pocket liability for the patient.

The NO_INSURANCE category accounts for 8,807 of these encounters, but a substantial portion of zero-coverage visits also appear under Medicaid and other payers, suggesting billing or eligibility gaps beyond just uninsured patients.

Average Claim Cost by Payer

Medicaid patients carried the highest average claim cost at \$6,205 per encounter, followed by NO_INSURANCE patients at \$5,593. Dual Eligible patients — those covered by both Medicare and Medicaid — had the lowest average at \$1,696, reflecting supplemental coverage reducing net claims.

Payer	Encounters	Avg Claim Cost
Medicaid	1,443	\$6,205
NO_INSURANCE	8,807	\$5,593
Anthem	704	\$4,237
Humana	1,084	\$3,269
Blue Cross Blue Shield	925	\$3,246
Medicare	11,371	\$2,168
Dual Eligible	912	\$1,696

Top Procedures by Frequency

Assessment of health and social care needs is the most frequently performed procedure at 4,596 instances, followed by Hospice care at 4,098 and Depression screening at 3,614. The prevalence of mental health and palliative care procedures reflects the senior-heavy patient demographic observed in the patient behavior section.

4. Patient Behavior Analysis

Demographics

The patient base is heavily skewed toward older adults. Patients aged 65 and above account for 65% of the total patient population — 633 out of 974. This concentration in the senior segment has direct implications for the types of procedures performed, the frequency of encounters, and the overall cost profile of the hospital.

Age Group	Patients	% of Total
65+ (Senior)	633	65.0%
50-64 (Older Adult)	175	18.0%
35-49 (Middle Adult)	157	16.1%
18-34 (Young Adult)	9	0.9%

Encounter Frequency

Across the 11-year dataset, 974 unique patients generated 27,891 encounters — an average of 29 encounters per patient. This high revisit rate suggests a chronically ill patient population with ongoing treatment needs rather than one-off visits. The gender split is near equal at 494 male and 480 female patients.

30-Day Readmissions

The 30-day readmission analysis identified 973 patients who returned to the hospital within 30 days of a previous encounter. Given the senior-heavy demographic and the prevalence of chronic conditions such as congestive heart failure and renal disease observed in the encounter data, this readmission rate is consistent with expectations for this patient profile.

Readmission patterns were identified using a self-join approach on the encounters table in MySQL, flagging any encounter that occurred within 30 days of the stop date of a prior encounter for the same patient.

5. Key Findings Summary

#	Area	Finding
1	Encounters	Volume peaked in 2014 at 3,885 encounters and again in 2021 at 3,500.
2	Encounters	Ambulatory visits dominate at 44.9%. Emergency and inpatient combined account for just 12.4%.
3	Encounters	95.9% of all encounters lasted under 24 hours. Longer stays are concentrated in inpatient and ICU classes.
4	Cost & Coverage	48.7% of encounters — 13,586 out of 27,891 — had zero payer coverage.
5	Cost & Coverage	Medicaid patients had the highest average claim cost at \$6,205, followed by NO_INSURANCE at \$5,593.
6	Cost & Coverage	Assessment and depression screening procedures are the most frequently performed, reflecting the chronic care profile of the hospital.
7	Patient Behavior	65+ seniors make up 65% of all patients — the hospital serves a predominantly elderly population.
8	Patient Behavior	Average of 29 encounters per patient across the dataset period, indicating high revisit rates.
9	Patient Behavior	973 patients were readmitted within 30 days, consistent with the chronic illness profile of the patient base.

6. Tools & Methodology

This project followed a structured four-phase workflow:

- Excel — Data profiling across all five tables, pivot-based summary analysis, and key finding documentation. All summary values are formula-driven.
- Python (Pandas, NumPy) — Exploratory analysis covering readmission logic, age group segmentation, cost percentile breakdowns, and correlation analysis. No visualisations were produced at this stage as all charts were reserved for Power BI.
- MySQL — Nine structured queries across three objectives, covering window functions, CTEs, self joins, DATEDIFF-based readmission logic, and aggregation with HAVING and LIMIT clauses.
- Power BI — Three-page interactive dashboard with KPI cards, slicers for year, payer, and encounter class, and page navigation across all three sections.

End of Report